

졸업프로젝트 면담

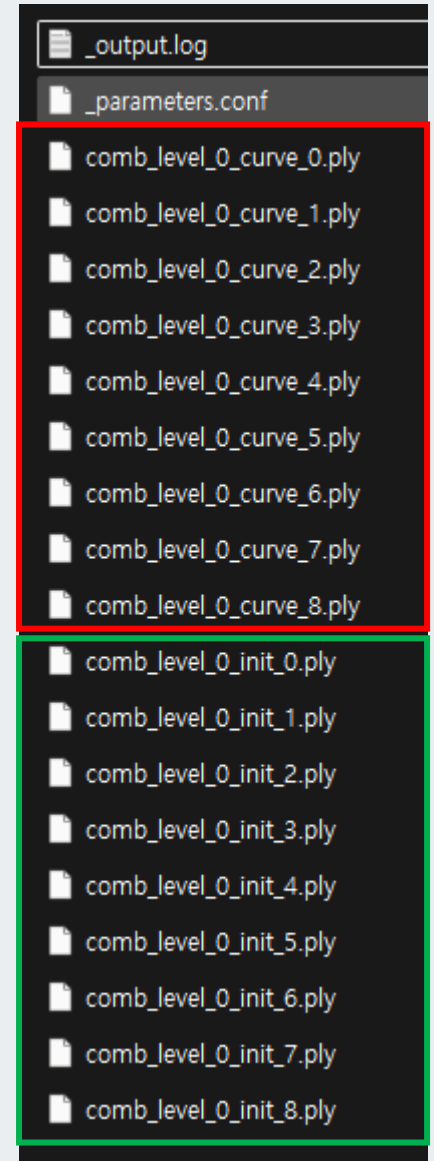
2026.04.08.(수)

Baseline Results

No.	Build Type	Input Dataset	Output	Runtime
1-1	Augmented Surface Interpolation	Humanoid_rm (vertex 2,000)	[level 0] init.ply 9개, curve.ply 9개	2시간 7분
1-2		Humanoid_new (vertex 2,818)	[level 0] init.ply 9개, curve.ply 9개	8분
2		Sphere Through Hole (vertex 2,500)	[level 0] init.ply 5개, curve.ply 5개 [level 1] init.ply 9개, curve.ply 9개	96분 (= lev_1 50분 + lev_2 46분)
3	Augmented Surface Extrapolation	Leaf	comb_exp__ 0 .ply ~ comb_exp_122.ply	[조기 중단] 122/450 steps : 2시간 17분
4		Head Punch	comb_exp_0.ply ~ comb_exp_32.ply	37분
5	Elastic Surface Extrapolation	Leaf	comb_exp_0.ply ~ comb_exp_450.ply	1시간 15분
6		Head Punch	comb_exp_0.ply ~ comb_exp_32.ply	3분
7	Repulsive Means	Mean_X	init.ply, input.ply 3개, mean.ply 6개, tpe_mean.ply 6개	1시간 38분
8	Shape From Metric	Hyperbolic Disk	에러나서 실행 실패	
9		Torus	undeformed.ply, deformed.ply	1시간 27분

Augmented Surface Interpolation

- Level의 역할: Multi-level Optimization
 - 저해상도에서 고해상도로
 - `comb_level_[L]_curve_[K]`
- **init** 파일과 **curve** 파일
 - **init**: 최적화를 시작하기 전 초기 경로를 저장한 파일로 상태를 확인하는 용도
 - **curve**: 최적화가 완료된 결과물
 - `level_[L]_curve` 파일은 `level_[L+1]_init` 파일로 사용됨



Augmented Surface Extrapolation

- Runtime:

	Leaf	Head Punch
Scary TPE	약 75초	약 75초
Spooky + Adaptive TPE	약 180초	약 300초
Spooky TPE	약 65초	약 90초

```
1 // 3가지 TPE
2 // SCARY : 정확도 높음, 속도 느림
3 // SPOOKY : 정확도 중간, 속도 빠름
4 // Adaptive : 정확도 높음, 속도 중간
5 ObjectiveFunctional<DefaultConfigurator> *chosenTPE = nullptr;
6
7 if ( Config.TPE.Type == SCARY ) {
8     chosenTPE = &scTPE; // ScaryTPE
9 }
10 else if ( Config.TPE.Type == SPOOKY ) {
11     if ( Config.TPE.useAdaptivity ) {
12         chosenTPE = &adspTPE; // AdaptiveFastMultipoleTPE
13     }
14     else {
15         chosenTPE = &spTPE; // SpookyTPE
16     }
17 }
```

Augmented vs. Elastic Surface Extrapolation



```
1 Exp2Energy<DefaultConfigurator> Felast( W, s0, s1 );
2 DifferenceExp2Energy<DefaultConfigurator> Ftpe( TPE, TPG, s0, s1 );
3 BarycenterExp2Energy<DefaultConfigurator> Fbary( s0, s1, nonDirichletIndices );
4 DirichletExp2Energy<DefaultConfigurator> Fdir( s0, s1, dirichletIndices );
5
6 VectorType Weights( 4 );
7 Weights << Config.Energy.elasticWeight, Config.Energy.tpeWeight, Config.Energy.dirichletWeight,
8           Config.Energy.barycenterWeight;
9
10 AdditionGradient<DefaultConfigurator> F( Weights, Felast, Ftpe, Fdir, Fbary );
11
12 CombinedExp2Gradient<DefaultConfigurator> DF( W, TPE, TPG, s0, s1,
13         Config.dirichletVertices, nonDirichletIndices,
14         Config.Energy.elasticWeight,
15         Config.Energy.tpeWeight,
16         Config.Energy.dirichletWeight,
17         Config.Energy.barycenterWeight );
18
19 InverseCombinedExp2Gradient<DefaultConfigurator> invDF( W, TPE, TPG, s0, s1,
20         Config.dirichletVertices, nonDirichletIndices,
21         Config.Energy.elasticWeight,
22         Config.Energy.tpeWeight,
23         Config.Energy.dirichletWeight,
24         Config.Energy.barycenterWeight );
25
26 NewOpt::NewtonMethod<DefaultConfigurator> Solver( F, DF, invDF, Config.Optimization.maxNumIterations, 1e-8,
27         NEWTON_OPTIMAL, SHOW_ALL, 0.1,
28         Config.Optimization.minStepsize,
29         Config.Optimization.maxStepsize );
```

Augmented vs. Elastic Surface Extrapolation



```
1 Exp2Energy<DefaultConfigurator> F( W, s0, s1 );
2 Exp2Gradient<DefaultConfigurator> DF( W, s0, s1 );
3
4 NewtonMethod<DefaultConfigurator> Solver( F, DF, Config.Optimization.maxNumIterations, 1e-8,
5                                           NEWTON_OPTIMAL, SHOW_ALL, 0.1,
6                                           Config.Optimization.minStepsize,
7                                           Config.Optimization.maxStepsize );
8 Solver.setBoundaryMask( Config.dirichletVertices );
```

Question

1. [instant-meshes] Correspondence가 깨져서 실행되지 않음 → Rigging?
2. Dirichlet Vertices는 어떤 기준으로 설정하는지?
3. parameter 값들을 조정하면서 여러 환경을 테스트 or 다른 Mesh 데이터를 적용?